

Notes

Posterior mean genetic liability from family history

Model, sequential approximation, and evidence in `ltpred`

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<https://github.com/bvilhjal/ltpred> · v0.3.2

15 August 2026

Abstract

The LT-FH family of methods replaces a case/control label with $\mathbb{E}[a_i \mid \text{family data}]$ under a Gaussian liability-threshold model. For rectangular observations that is ordinary BLUP after the liabilities have been replaced by their truncated-normal means. LT-FH, LT-FH++ and ADuLT choose different rectangles; PA-FGRS also permits a censoring mixture of truncated laws. These are observation models, not different genetic models. These notes fix the estimand, the observation-model menu, the exactness boundary of the Pearson–Aitken (PA) approximation, and the sampling conditions under which a pedigree Haseman–Elston fit recovers h^2 . Simulation under the model’s own generative assumptions gives $\text{Corr}(\text{PA}, \text{Gibbs}) \geq 0.9977$ without the censoring mixture, a classic-LT-FH causal-SNP noncentrality ratio of $1.47 \pm 0.04 \times$ over case/control (independent SNPs), and a paired $+0.145 \pm 0.015$ LT-FH++ increment over matched ADuLT when onset is a deterministic function of liability and the CIP is known. The same personalised thresholds that supply that increment also remove a simulated cohort-confounding λ_{GC} of 16.46 ± 0.40 (Table 5). A PGS and μ_i remain complementary: joint test $R^2 = 0.339$ against 0.236 and 0.170 (Table 6). The GWAS and personalisation figures are upper bounds relative to register data: a pin-versus-interval ablation keeps almost all of the onset increment except at high prevalence, and pinning is calibrated only when onset is the CIP inverse of liability. They are not estimates of clinical utility.

The intended reader already uses GWAS, liability-scale h^2 , and Falconer’s threshold construction. The notes do not re-derive those. They are about what is being conditioned on, where the sequential approximation is exact, and which simulation claims survive their generative assumptions.

1 Estimand

Write a_i for the additive genetic liability of a designated proband, $\text{Var}(a_i) = h^2$ on the usual unit-liability scale, and ℓ_F for the vector of full liabilities on the people whose disease records are used (the proband may or may not be among them). Let D_F denote the complete observation model encoded by those records. The general target is

$$\mu_i = \mathbb{E}[a_i \mid D_F, A, h^2, K(\cdot)], \quad (1)$$

where A is the additive relationship matrix and $K(\cdot)$ is the prevalence or cumulative-incidence model that turns status and age into interval endpoints or mixture weights. For LT-FH, LT-FH++ and ADuLT, D_F is the rectangular event $\ell_F \in C_F$, one interval per person. The PA-FGRS censoring mixture is a mixture of truncated laws and is not, in general, one rectangle.

If ℓ_F were observed continuously, (1) would be the selection index / animal-model BLUP

$$\mathbb{E}[a_i | \ell_F] = \text{Cov}(a_i, \ell_F) \text{Var}(\ell_F)^{-1} \ell_F \quad (2)$$

[4, 5]. A single relative with relationship r contributes $r h^2 \ell_r$. Disease records do not give ℓ_F . They give $\ell_F \in C_F$. Under joint normality,

$$\mathbb{E}[a_i | \ell_F \in C_F] = \text{Cov}(a_i, \ell_F) \text{Var}(\ell_F)^{-1} \mathbb{E}[\ell_F | \ell_F \in C_F]. \quad (3)$$

The BLUP weights are unchanged; only the right-hand side is replaced by a truncated-normal mean. The map from the binary (or censored) records to μ_i is therefore nonlinear, and reduces to (2) only when every interval collapses to a point.

Two immediate consequences, both easy to get wrong in an analysis.

1. **The proband's own diagnosis is optional.** Include it when μ_i is a GWAS phenotype *constructed from* that diagnosis [7, 18]. Leave the corresponding interval as $(-\infty, \infty)$ when the same diagnosis is the outcome you will later predict or classify. Conditioning on the answer is leakage, not sophistication.
2. **(1) is not a SNP polygenic score.** No marker effects enter. Combining μ_i with a PGS is a downstream model [8, 13].

The multivariate threshold model on a pedigree is Sham's framework [21]. So, Kwan, Cherny and Sham [22] already combined family history with known loci, age, and competing mortality; they targeted absolute risk. Here the target is the breeding value, used as a phenotype.



Figure 1: The published methods differ in how they build D_F . Rectangular methods share (3); PA-FGRS may instead use the mixture in (10).

2 Covariance

The liability split is the usual one,

$$\ell = a + e, \quad a \sim \mathcal{N}(0, h^2), \quad e \sim \mathcal{N}(0, 1 - h^2), \quad a \perp\!\!\!\perp e, \quad (4)$$

so $\ell \sim \mathcal{N}(0, 1)$ and a case is $\ell > T$ with $T = \Phi^{-1}(1 - K)$ at a single prevalence K [23, 3]. On a pedigree,

$$\text{Cov}(a_i, a_j) = h^2 A_{ij}, \quad \text{Cov}(\ell_i, \ell_j) = h^2 A_{ij} \ (i \neq j), \quad \text{Var}(\ell_i) = 1, \quad (5)$$

with $A_{ij} = 2\phi_{ij}$. Mates are taken as unrelated ($A_{mf} = 0$): the default A is not a generative model of assortment. Inbreeding is handled by the tabular method [6], after which full liabilities are rescaled to unit variance so that $\Phi^{-1}(1 - K)$ still means prevalence K .

Shared environment is optional and off-diagonal,

$$\text{Cov}(\ell_i, \ell_j) = h^2 A_{ij} + c^2 C_{ij} + m^2 M_{ij}, \quad h^2 + c^2 + m^2 \leq 1. \quad (6)$$

The target a_i still couples to relatives only through $h^2 A$. Identification is the usual contrast arithmetic: C (full-sib environment) is the excess of sib covariance over parent–offspring (both have $A = \frac{1}{2}$); M (couple) is the covariance of genetically unrelated mates, and therefore confounds household sharing with assortment [20]. A directional maternal effect is not a symmetric kernel and is not M . An environment that scales with A is absorbed into h^2 and is not estimable.

Nuclear families give roughly three contrasts; a bank of relationship-specific environments needs the relative types that extended pedigrees supply. This is the pedigree analogue of why ACE is identified from MZ/DZ twins and $A + C + D$ is not.

One inherited LTFHPlus convention is worth knowing before comparing constructors: two same-side half-sibs in the role grammar are related $\frac{1}{2}h^2$ to *each other* (implied shared second parent). A pedigree in which they have distinct other parents must be built from A , not from those role labels.

3 The observation-model menu

LT-FH, LT-FH++ and ADuLT are names for rectangular D_F , not for an inference algorithm. Gibbs and PA both apply to the first three. PA-FGRS, in the published specification and in the implementation, is PA-specific.

Classic LT-FH [7]. One K , one T . Cases $[T, \infty)$, controls $(-\infty, T]$, relatives included. This is the configuration-specific successor of GWAX [15].

LT-FH++ and ADuLT [18, 19]. Age, sex and cohort enter only through a CIP $K(t; s, b)$ and

$$T_i = \Phi^{-1}(1 - K(t_i; s_i, b_i)). \quad (7)$$

A control at current age c_i is truncated above at $T_i(c_i)$. A case with onset a_i is *pinned* at $T_i(a_i)$ (lower = upper). The same T_i with relatives is LT-FH++; without them, ADuLT.

Pinning is the strong assumption in this construction, and it is easy to understate. It says that the CIP is an error-free map from liability to onset, that onset dates are exact, and that two cases with the same stratum and onset have identical liability—zero residual variance given a_i . The inverse map is then the identity: the observation *is* the liability. When onset is recorded in a window, an exact likelihood must integrate over the recording bin under an explicit onset model. An interval $[T_i(a_i), \infty)$ or the lifetime interval $[T_{\text{pop}}, \infty)$ is a conservative partial-information encoding, not that general likelihood. On the same simulated families, the interval keeps almost all of the onset increment except at $K = 0.30$, where the pin adds another 0.01–0.02 in the squared-correlation proxy. When onset only tends to track liability ($\rho = 0.6$), the pin over-conditions (slope 0.92 under heavy censoring).

Because the BLUP weights in (3) do not depend on the T_i , a shared threshold error is mostly a calibration shift. A stratum-specific error also reorders scores.

Base PA-FGRS [12]. Observed cases use the *lifetime* interval $[\Phi^{-1}(1 - K_{\text{pop}}), \infty)$. Age enters for censored controls through the mixture in Section 4.3, not through the case threshold. Helpers that put cases above $T_i(a_i)$ are a different observation model; they are not PA-FGRS_{ADT}. Kendler’s FGRS [10] is a third thing and is not implemented.

PGS and μ_i . Split $a = s + u$ with $\text{Var}(s) = p := h_{\text{SNP}}^2/h^2$. If two standardised scores satisfy $\text{PGS} \perp\!\!\!\perp \mu \mid a$, then

$$\text{Corr}(\text{PGS}, \mu) = a_{\text{PGS}} b_{\mu} \sqrt{p} = \sqrt{R_{\text{PGS}}^2 R_{\mu}^2 p}. \quad (8)$$

The scores can each be genuinely predictive of a and still almost uncorrelated, because each is mostly noise relative to a [8, 13]. Non-overlapping discovery and family samples remove one shared noise term. They do not imply the residual is zero (ancestry, assortment, indirect effects, selection).

4 Computing the truncated mean

There is no closed form for $\mathbb{E}[\ell_F \mid C_F]$ once several intervals are live. Two approximations are used.

4.1 Gibbs

Coordinate-wise inverse-CDF sampling of the conditional normals [11], with coefficients from the precision $Q = \Sigma^{-1}$: $\text{sd}_j = \sqrt{1/Q_{jj}}$, $P_{ij} = -Q_{ij}/Q_{jj}$. Pinned coordinates are held. Far-tail intervals cannot be drawn on the probability scale (Φ underflows near $|z| \approx 38.5$); those use a log-scale Rayleigh construction. Σ must be strictly positive definite.

The Monte Carlo standard error is the batch-means estimator [9] that LTFHPlus takes from `batchmeans`. Families that share a structure share one factorisation of Σ ; the rest is embarrassingly parallel. This is the sampling-based reference for every no-mixture claim below.

4.2 Pearson–Aitken

If one coordinate of a multivariate normal is selected from $\mathcal{N}(m_i, v_i)$ to (m^*, v^*) , the others update by [17, 1]

$$m_j \leftarrow m_j + \frac{\Sigma_{ji}}{v_i}(m^* - m_i), \quad \Sigma_{jk} \leftarrow \Sigma_{jk} + \frac{\Sigma_{ji}\Sigma_{ik}}{v_i^2}(v^* - v_i). \quad (9)$$

Mendell and Elston applied this sequentially to threshold traits [16]. The implementation folds observed members last-to-first, then applies the target’s own interval to the updated $\mathcal{N}(m_0, v_0)$. For the genetic target the last step is a no-op (bounds $(-\infty, \infty)$). For $\mathbb{E}[\ell_o \mid C_F]$ it is the same estimand Gibbs uses.

After one truncation the selected law is no longer Gaussian. PA keeps the first two moments and proceeds as if it were. Hence: exact for a single interval or a pin; a sequential two-moment approximation thereafter. Fold order changes the error. Canonicalising to a sorted role set makes the result a function of the pedigree shape, not of input row order—reproducibility, not accuracy.

On the no-mixture structures we have tested, that approximation error is small enough to ignore for the posterior *mean* (Section 7). It has not been characterised for the mixture, and PA’s reported conditional variance is itself an approximation. A PA standard error of zero means no Monte Carlo error.

4.3 Censored-control mixture

A person unaffected only up to current follow-up is a mixture of a lifetime control and a not-yet-onset future case [12]. With current CIP K_i and lifetime K_{pop} ,

$$\pi = \frac{\Phi_{\text{below}}}{\Phi_{\text{below}} + (1 - \Phi_{\text{below}})(K_{\text{pop}} - K_i)/K_{\text{pop}}}, \quad (10)$$

where the split is the *lifetime* threshold and Φ_{below} is the current conditional mass below it. The selected moments are the corresponding two-component mean and variance. Age enters only through K_i .

The factor $(K_{\text{pop}} - K_i)/K_{\text{pop}}$ is the CIP probability that a future case has not yet onset. That is an onset-timing assumption: among people who will eventually be cases, time-to-onset is independent of liability, and censoring is non-informative given the stratum. If higher liability advances onset, the not-yet-onset component is not the above-threshold tail assigned to it. A PA generative check at $\rho = 0.6$ finds statistically detectable but practically negligible ranking shifts and the usual downward slope shift; the pin, not the mixture, is what breaks materially (Section 7). There is still no Gibbs implementation of (10).

5 Implementation

The mathematics above is small dense linear algebra on one family. The software problem is many independent families, most of which share a pedigree shape.

Grouping. Families with the same ordered role set share one Σ . Both engines group on that key, factorise once, and apply the per-family intervals in a compiled kernel. The object API (**Family/Member**) does this grouping internally. The array API receives already-aligned bounds and is the hot path.

Gibbs. The inner sweep is Numba-JIT'd and **prange**-parallel across families in a group. Batch-means summaries—running sums and sums of squares—are streamed, so Monte-Carlo-SE memory is $O(F)$ rather than $O(n_{\text{sim}}F)$, and the posterior variance is the same streaming second moment. Each family carries its own RNG seed: the draw does not depend on which worker picked it up. Without Numba the identical code runs serially. Far-tail inverse-CDF draws switch to a log-scale Rayleigh construction once Φ underflows.

PA. The same grouping applies. Fold order is canonicalised to a sorted role set so the result is a function of the pedigree shape, not of input row order—reproducibility, not a claim of smaller approximation error (Section 4.2).

What the timings measure. Section 7 quotes a 203–492 $\times$ object-path speed-up of PA over grouped Gibbs at four threads. That includes Python-side assembly. The array path, on already-aligned inputs, ran at 1.2–3.8 million families per second in the same warmed run—another 21–52 $\times$ over the object path. Those array rates are a hot-kernel measurement, not end-to-end register ingestion.

Two different uncertainties. Write $\hat{\mu}_i$ for the reported estimate of (1). Then **var** is $\text{Var}(a_i | D_F)$, the posterior spread. It does not shrink with more Gibbs draws. **se** is the estimator's own error in $\hat{\mu}_i$: batch-means Monte Carlo for Gibbs, identically zero for PA. A PA standard error of zero means no Monte Carlo error, not zero approximation error, and not zero posterior variance. On a seven-observation fold—the regime where PA **var** is itself an approximation—the sequential-moment variance sits within 5% of the Gibbs Monte-Carlo variance.

What is not implemented. There is no Gibbs version of (10). Multi-trait estimation is Gibbs-only. The shipped environment kernels C and M feed back into single-trait prediction through c^2 and m^2 ; a directional maternal effect is not M .

6 Fitting h^2 is a different problem

Prediction *conditions* on h^2 . Bring a liability-scale estimate; if what you have is observed-scale (OLS / GREML / LDSC on 0/1), Lee et al. [14] give

$$h_{\text{liab}}^2 = h_{\text{obs}}^2 \cdot \frac{K(1-K)}{z^2} \cdot \frac{K(1-K)}{P(1-P)}, \quad z = \phi(\Phi^{-1}(1-K)), \quad (11)$$

with P the sample case fraction ($P = K$ drops the second factor).

A pedigree fit of h^2 from the same family records is available, but it is not “LT-FH++ with h^2 free.” The implementation is data augmentation: draw $\ell | C_F, \Sigma(h^2)$, then a damped Haseman–Elston step

$$\widehat{h^2} = \frac{\sum_{\text{pairs}} A_{ij} \ell_i \ell_j}{\sum_{\text{pairs}} A_{ij}^2}. \quad (12)$$

This is a stochastic-approximation fixed point, not a posterior. The number it reports as a standard error is within-chain Monte Carlo noise. Across-cohort SD is 20–30× larger (Table 7).

Law of total expectation recovers $\mathbb{E}[\ell_i \ell_j] = h^2 A_{ij}$ from the augmented products only for independent, non-overlapping families under the declared sampling contract and a *common* T per trait. Personalised or onset-pinned rectangles—even the internally coherent ones that prediction wants—are outside that identified moment design and are refused. This is an explicit contract, motivated by exploratory failures, not a current quantitative benchmark for personalised fitting. Ascertainment is not cured by more sweeps. The damage is severe and silent: on families selected through an affected proband with a *true* h^2 of zero, (12) returns $h^2 = 1.0$, and a realised case share of 6.4% against an asserted 5.0% (1.27×) already inflates h^2 by +0.230 (Section 7.9). The claim is therefore screened for marginal inconsistency rather than taken entirely on trust: the supplied T implies $K = 1 - \Phi(T)$, under population sampling each role’s case count is Binomial(n_{fam}, K), and a gross departure is refused. Passing this role-wise screen does not certify the joint family-pattern distribution. The same remarks apply to the multiple HE fit of (h^2, c^2, m^2) , where saturation shows up as an exhausted residual rather than a component at the clamp. A family bootstrap, when clusters are independent, estimates sampling variability under the fitted design. It does not de-bias selection.

Reweighting, where it is available. The augmentation of a *given* family with *given* statuses is already the correct conditional law; what selection corrupts is the *mix* of families. So when the inclusion probability π_f is known by design and strictly positive for every stratum, weighting each family’s contribution by $w_f = 1/\pi_f$ in both the numerator and the denominator of (12) re-mixes the sample to population proportions and restores the estimator. Two conditions bind. *Positivity*: a design that samples no family from some stratum has $\pi = 0$ there, and no weight reconstructs what was never observed—proband ascertainment is exactly this case. Selection on family history is IPW-correctable only if every complete observed family pattern has a known, strictly positive inclusion probability. The weighted role-marginal screen can falsify some wrong weight sets; it is not a general positivity test. *Efficiency*: at $K = 0.05$ a 50/50 case/control cohort needs weights up to 19, so the effective sample size falls well below the nominal one.

A multiplicative observed-to-liability factor such as (11) cannot substitute for this. It is a function of (K, P) alone, whereas at fixed K a true h^2 of 0.5 and of 0 both drive (12) to 1.0; no invertible constant maps both back. It also transforms an *observed-scale* estimate, and (12) is already on the liability scale. If the question is h^2 from an ascertained case/control GWAS of unrelated individuals, use LDSC or GREML and (11). If the question is h^2 from population-sampled nuclear families with one threshold, the pedigree HE is a reasonable moment estimator and is calibrated in that design (Section 7). Those are not the same analysis. The benchmark’s “Lee arm” is only a diagnostic: it standardises each family role by its sample case rate, pools multi-relative HE moments, and applies the proband’s sample fraction in (11). It is not an implementation or empirical evaluation of conventional unrelated-sample LDSC or GREML.

7 Evidence

Unless noted, $\pm$ is the SE of a mean across independent simulated cohorts. The numbers below are from committed benchmark artifacts (campaigns of 31 July, 1 August and 10–14 August 2026). They do not automatically validate later code. Every GWAS panel uses independent SNPs; a real-LD run has not been done. There is no locked comparison to R LTFHPlus. Table 1 is the load-bearing subset; the rest of this section says what each number is allowed to mean.

Two ratios appear and are not interchangeable. An *NCP ratio* is $(\bar{\chi}^2_{\text{causal}} - 1)$ over the same quantity for case/control. An *eff-N proxy* is a squared-correlation ratio to the case/control label.

Table 1: Load-bearing simulation claims. $\pm$ is a replicate SE except where marked as a 95% CI half-width. Independent SNPs throughout; no row uses the censoring mixture (10).

claim	number	design
PA-Gibbs agreement	0.9977–0.9999	27-cell classic LT-FH grid, no mixture
Object-path speed-up	203–492 $\times$	4 threads, this machine
Classic LT-FH NCP	1.47 $\pm$ 0.04 $\times$	10,000 probands, 30 causal SNPs
LT-FH++ minus ADuLT	+0.145 $\pm$ 0.015	paired 95% CI; known CIP, pinned onset
Cohort-blind λ_{GC}	16.46 $\pm$ 0.40	4 $\times$ prevalence trend / 30 y
PGS + LT-FH test R^2	0.339 vs 0.236, 0.170	held-out genetic liability; $p = 1$ by construction
IPW 50/50 fitted h^2	0.495 (from 1.000)	$N = 10,000$, $K = 0.05$, truth 0.5

7.1 The sequential approximation

On a 27-cell grid of classic LT-FH rectangles (three family structures $\times$ three $h^2 \times$ three K ; five cohorts of 1,500 families), $\text{Corr}(\text{PA}, \text{Gibbs})$ is 0.9977–0.9999. Stress pedigrees remain ≥ 0.9984 . Table 2 is the $h^2 = 0.5$, $K = 0.05$ slice. Calibration slopes of correctly specified PA sit on 1 to within simulation error (e.g. 1.006 ± 0.008 at $K = 0.05$, parents + two sibs). That is the argument for using PA as the default engine when the mixture is off. It is not a licence to treat PA and Gibbs as interchangeable once (10) is on.

At four threads on one machine the object-path PA sweep is 203–492 $\times$ faster than grouped Gibbs (2,000 families, 25,000 draws). The ratio moves with thread count because Gibbs is the parallel engine. The useful statement is that the approximation is cheap enough for biobank-scale C_F , not a hardware-independent constant.

Table 2: PA versus Gibbs, classic LT-FH bounds, $h^2 = 0.5$, $K = 0.05$, five seeds. Correlation is with simulated a_i . The eff- N proxy is not an NCP ratio.

Structure	corr. Gibbs	corr. PA	eff- N / c-c	$\text{Corr}(\text{PA}, \text{Gibbs})$
parents	0.382 $\pm$ 0.009	0.382 $\pm$ 0.009	1.36 $\pm$ 0.03 $\times$	0.9997
parents + 2 sibs	0.437 $\pm$ 0.009	0.437 $\pm$ 0.009	1.66 $\pm$ 0.07 $\times$	0.9997
extended	0.419 $\pm$ 0.014	0.420 $\pm$ 0.014	1.71 $\pm$ 0.09 $\times$	0.9997

7.2 Classic LT-FH as a GWAS phenotype

Three genotype/effect/cohort replicates; 10,000 probands, 5,000 independent SNPs, 30 causal, $h^2 = 0.5$, $K = 0.05$, parents plus one sibling (Table 3). The same classic LT-FH C_F , inferred by either engine, gives an NCP ratio of $1.47 \pm 0.04 \times$ over case/control, with λ_{GC} consistent with 1. A single-seed 1.53 $\times$ does not survive replication. The oracle (true a_i) is 8.8 $\times$: most of the liability is still unused, as it should be when $h^2 = 0.5$ and relatives are few.

Table 3: Independent-SNP GWAS. NCP uses $\overline{\chi^2} - 1$. Power is the fraction of causal SNPs at $p < 5 \times 10^{-8}$.

phenotype	mean causal χ^2	causal-SNP NCP ratio / c-c	power at 5e-8	λ_{GC}
case/control	39.63 $\pm$ 2.19	1.00 $\times$	41.1 $\pm$ 2.2%	1.011 $\pm$ 0.009
classic LT-FH (Gibbs)	57.53 $\pm$ 1.87	1.468 $\pm$ 0.040 $\times$	48.9 $\pm$ 4.0%	1.012 $\pm$ 0.019
classic LT-FH (PA)	57.60 $\pm$ 1.91	1.469 $\pm$ 0.039 $\times$	48.9 $\pm$ 4.0%	1.007 $\pm$ 0.021
oracle true g	337.37 $\pm$ 2.07	8.771 $\pm$ 0.573 $\times$	75.6 $\pm$ 1.1%	1.047 $\pm$ 0.015

7.3 What personalisation is worth

The integrated LT-FH++ panel uses age/sex/cohort CIP, coherent follow-up, competing death, ascertainment, and stratified null SNPs (Table 4). Onset is the CIP inverse of true ℓ , and cases are pinned at that T_i —so a case’s full liability is essentially observed. Ten paired replicates. Accuracy, slope and NCP are adjusted for proband sex and birth year.

Table 4: Integrated personalised LT-FH++ genotype GWAS, ten paired replicates. NCP is $(\chi^2 - 1)$ over case/control. The last column is stratified-null λ_{GC} before and after the same covariate adjustment.

phenotype	adjusted corr	adjusted slope	adjusted causal-SNP NCP ratio / c-c	stratified-null λ raw $\rightarrow$ adjusted
case/control	0.586 ± 0.008	1.123 ± 0.021	$1.000 \pm 0.000\times$	$1.140 \pm 0.051 \rightarrow 1.019 \pm 0.024$
ADuLT (full personalised proband CIP, no FH)	0.600 ± 0.008	1.004 ± 0.018	$1.049 \pm 0.004\times$	$1.024 \pm 0.047 \rightarrow 1.014 \pm 0.031$
LT-FH single-K	0.629 ± 0.008	1.183 ± 0.020	$1.149 \pm 0.007\times$	$1.131 \pm 0.046 \rightarrow 0.990 \pm 0.041$
FH + age CIP (ablation)	0.639 ± 0.008	1.009 ± 0.016	$1.189 \pm 0.006\times$	$0.999 \pm 0.033 \rightarrow 1.018 \pm 0.041$
FH + age + sex CIP (ablation)	0.639 ± 0.008	1.008 ± 0.016	$1.188 \pm 0.006\times$	$0.985 \pm 0.032 \rightarrow 1.011 \pm 0.036$
FH + age + cohort CIP (ablation)	0.640 ± 0.008	0.995 ± 0.015	$1.194 \pm 0.006\times$	$1.014 \pm 0.034 \rightarrow 1.020 \pm 0.034$
LT-FH++ (full age + sex + cohort CIP)	0.640 ± 0.008	0.995 ± 0.015	$1.194 \pm 0.006\times$	$0.996 \pm 0.036 \rightarrow 1.010 \pm 0.039$

The paired LT-FH++ minus ADuLT increment is $+0.145 \pm 0.015$ NCP-ratio units (95% CI half-width). Age accounts for most of the step beyond classic LT-FH ($+0.040 \pm 0.008$); cohort adds a further $+0.006 \pm 0.002$; sex adds nothing to adjusted NCP (-0.0004 ± 0.0010). A prespecified sex-only panel nevertheless closes a female–minus–male mean-score-error gap of 0.051 ± 0.001 . Sex-specific CIP is a calibration device in this design, not a proven power device.

Read Table 4 as: under a known CIP and a deterministic onset map, family history still adds association signal after the proband’s own T_i is personalised, and most of that increment is already present once age is in the CIP. Do not read it as the increment you will see in a register, where onset is noisy and $K(t; s, b)$ is estimated. The pin-equals-liability identity is an assumption of the observation model, and the simulation honours it. A separate, no-SNP ablation on the same families (section 3 of the benchmark report) finds that replacing the pin with the interval $[T(\text{onset}), \infty)$ keeps almost all of the onset increment except at high prevalence, where the pin adds another 0.01–0.02 in the squared-correlation proxy. Most of what age of onset contributes is the lower bound, not the point mass.

7.4 Cohort confounding

Because the BLUP weights in (3) do not depend on the T_i , a *shared* threshold error is mostly a calibration shift. A *stratum-specific* error also reorders scores, and—if the stratum is correlated with a marker—induces association on that marker. Table 5 is the designed case: a secular prevalence trend plus null SNPs correlated with birth year.

Table 5: Stratified-null λ_{GC} under a lifetime prevalence trend of R per 30 birth years. Three replicates. Truly independent null SNPs stay near 1 for every method.

prevalence trend R per 30 y	FH + cohort-specific K	single-K FH	case/control
1	0.960 ± 0.012	0.960 ± 0.012	0.899 ± 0.034
2	0.964 ± 0.054	4.539 ± 0.253	2.008 ± 0.257
3	0.922 ± 0.057	10.275 ± 0.078	4.616 ± 0.220
4	1.007 ± 0.040	16.460 ± 0.399	7.666 ± 0.094

At $R = 1$ there is no trend-driven inflation. At $R = 4$, cohort-blind family thresholds reach $\lambda_{GC} = 16.46 \pm 0.40$; the same family model with cohort-specific K stays at 1.007 ± 0.040 . Case/control is also inflated, just less so. This is the argument for treating personalised CIP as a confounding-control device, not only as a power tweak. It is an independent-SNP illustration of the cohort component; it is not the full age/sex/cohort LT-FH++ design of Table 4.

7.5 Calibration, and a wrong h^2

A correctly specified posterior mean is self-calibrating: the regression of true a_i on $\hat{\mu}_i$ has slope 1. On the classic-LT-FH grid that slope sits on 1 to within simulation error (e.g. 1.006 ± 0.008 at $K = 0.05$, parents + two sibs). One replicated departure remains: at $K = 0.01$ with an extended pedigree the top-decile realised/predicted ratio is 0.946 ± 0.018 , about 3 SE below 1.

A wrong assumed h^2 leaves the ranking almost untouched and tilts the scale. With true $h^2 = 0.5$ and $K = 0.05$, assumed h^2 from 0.2 to 0.8 sweeps the slope from 2.240 ± 0.015 down to 0.678 ± 0.006 while $\text{Corr}(\hat{\mu}, a)$ stays in 0.429–0.431 ($\text{SE} \leq 0.004$). Use $\hat{\mu}$ as a GWAS phenotype or a ranker under a roughly right h^2 ; do not treat its numerical scale as $\mathbb{E}[a_i \mid C_F]$ unless h^2 is the one the score was computed at.

Heavy tails, generative assortment at phenotypic $\rho \approx 0.3$, and an unmodelled sibship $c^2 = 0.15$ each move the slope by only 0.01–0.03. The ranking cost is small but not uniformly negligible: assortment costs 0.007 and sibship 0.005 in correlation, while heavy tails cost 0.016 (0.4983 ± 0.0045 against 0.4820 ± 0.0024), about 3% of the control correlation and roughly 3 SE. A factor-of-two error in prevalence moves the slope to 0.89 or 1.13. Threshold provenance is the calibration hazard; the Gaussian family model is not.

7.6 Complementarity with a PGS

Five replicates, 10,000 probands, 2,000 independent SNPs, 50/50 train/test, PGS trained on the train case/control GWAS (infinitesimal weights; no LD to shrink). Table 6 is scored against held-out a_i .

Table 6: PGS and classic LT-FH on a held-out test half. R^2 is against true a_i . The joint model is OLS on the two scores.

score	corr with held-out g	R^2
case/control label	0.343 ± 0.007	0.118 ± 0.005
PGS	0.485 ± 0.009	0.236 ± 0.009
LT-FH (PA)	0.413 ± 0.003	0.170 ± 0.003
PGS + LT-FH joint (OLS)	—	0.339 ± 0.009

Incremental R^2 : $+0.104 \pm 0.007$ for LT-FH over PGS, $+0.169 \pm 0.019$ for PGS over LT-FH (95% CI half-widths). Observed $\text{Corr}(\text{PGS}, \mu) = 0.2009 \pm 0.0046$ against (8) at $p = 1$ (0.2003 ± 0.0050). Conditional independence holds by construction here. That is a verification of the identity, not a test of the failure modes. The portable number is the increment, not the absolute R^2 : 2,000 independent SNPs are a friendlier multiple-testing regime than a genome-wide panel.

7.7 The censoring mixture

The mixture (10) is PA-only. A generative check (five replicates of 20,000 nuclear families; mid-life and old-age censoring; three onset models) finds ranking costs that are resolvable but negligible in size: three of the ten paired ΔCorr contrasts exclude zero, the largest being -0.00034 with a 95% CI half-width of 0.00011, and all ten CIs are tighter than ± 0.0002 . A cost of 3×10^{-4} in correlation is statistically real and practically irrelevant; the honest statement is that the mixture does not meaningfully change ranking, not that no cost is detectable. Every Δslope CI excludes zero; the mixture always lowers the slope, by 0.007–0.025, and most strongly under heavy censoring.

The third onset model is a Gaussian copula of strength $\rho = 0.6$ between liability and the CIP quantile among lifetime cases. Higher liability then advances onset, with residual noise. Plain age truncation is no longer exact, and neither is the pin. Pinned slope falls to 0.92 under heavy censoring. The lifetime-interval slope sits at 1.13, between the crossing (1.20) and stochastic (1.03) extremes.

Case encoding still dominates calibration, but the right encoding depends on the onset model. Pin cases at their onset threshold when the crossing model is believed; use the lifetime interval when onset ages are unreliable or only partly liability-dependent. There is still no Gibbs implementation of (10), so this paragraph is not a PA–Gibbs comparison.

7.8 Moment recovery of h^2 and C

Twenty-five unascertained population cohorts, 3,000 families, parents + two sibs, $K = 0.10$, one threshold (Table 7). Bias is small relative to across-cohort SD. The fitter’s Monte Carlo SE is not a sampling SE.

Table 7: Haseman–Elston data-augmentation fit of liability-scale h^2 under the design it is identified in.

true h^2	fitted mean	bias	empirical SD	reported MC SE	SD / MC SE
0.2	0.201	+0.001	0.037	0.0020	19×
0.4	0.387	-0.013	0.055	0.0022	25×
0.6	0.577	-0.023	0.063	0.0023	28×
0.8	0.786	-0.014	0.061	0.0026	24×

Joint $A + C$ on the same design (mean, emp. SD): true (0.4, 0.2) recovers 0.399 (0.057), 0.190 (0.034); true (0.6, 0) recovers 0.583 (0.057) and a boundary 0.013 (0.012). A point near zero is not a Type I rate. Omitting a real C is mostly a problem for interpreting h^2 (at true $c^2 = 0.3$, additive-only $\widehat{h^2} \approx 0.75$ versus ≈ 0.48 with C in the model), not for predicting a_i .

A research-only cross-trait HE, 25 cohorts, recovers $r_g = 0$ as -0.006 (SD 0.080) in the presence of phenotypic correlation 0.2, and $r_g = 0.6$ as 0.598 (SD 0.071).

7.9 What selection does to the same fit

Ten replicates, 10,000 analysed families, $K = 0.05$, true $h^2 = 0.5$, nuclear pedigrees. Each replicate draws a population, applies one selection rule, and fits the selected subset; only the rule differs between rows. `random_50` keeps half the population *independently of phenotype* through the identical code path, and is the control that separates selection-on-phenotype from the machinery.

Table 8: Fitted h^2 under selection, and the inverse-probability weighted refit. Truth is 0.5; † marks a design whose inclusion probability is zero for some stratum, where weighting is undefined.

selection	case share	unweighted $\widehat{h^2}$	IPW $\widehat{h^2}$
none (population)	0.050	0.523	—
random 50% (control)	0.049	0.506	—
50/50 case/control	0.499	1.000	0.495
20% case-enriched	0.199	1.000	0.473
affected proband	1.000	1.000	†
any affected member	0.293	1.000	†
affected relative only	0.000	1.000	†

Every phenotype-selected design in Table 8 is pinned at the clamp in every replicate. The sharper cell is $h^2 = 0$: the population arm returns 0.018 and the control 0.025, while all five selected designs return 1.000—complete heritability in a population with none.

It is bias, not variance, and not an unfinished chain. The gap is +0.500 at $N = 2,500$, 10,000 and 40,000 while the population arm’s shrinks toward zero; the same 0.9999 is reached from $h^2_{\text{init}} \in \{0.05, 0.5, 0.95\}$, so the iteration climbs to the ceiling from below rather than failing to leave a high start.

The clamp censors the magnitude. Evaluating (12) on the selected families’ own liabilities is uncensored and gives 1.68 under proband ascertainment, $3.4\times$ the truth. That statistic is a decomposition and *not* the mechanism: at $h^2 = 0$ it sits near zero while the fitter reports 1.0. The fitter never sees those liabilities. It sees truncated draws conditioned on the selected status pattern, so under proband ascertainment every augmented proband is redrawn above T , its relatives are pulled with it, and the cross-products stay positive whatever the truth.

7.10 Which relatives, and leaking the outcome

On five simulated three-generation populations, scoring every relative out to third degree rather than the named-role grammar (parents, full sibs, grandparents) raises $\text{Corr}(\hat{\mu}, a)$ from 0.498 ± 0.016 to 0.569 ± 0.013 (paired $+0.071 \pm 0.006$, 95% CI $[0.054, 0.089]$). The increment is the LT-FGRS point: the degree limit chooses *which* people enter C_F , not the kinship among those who do. An ancestral closure is required for that kinship to be exact.

For prospective prediction the proband’s own future diagnosis must not enter C_F (Section 1). On five synthetic registers the honest familywise-censored score has rank AUC 0.581 ± 0.031 for diagnosis in $(40, 70]$ after index age 40. Relatives’ post-index events raise AUC by $+0.065 \pm 0.014$ (they help discrimination; they are leakage for a prospective claim). The proband’s own future outcome raises AUC by $+0.204 \pm 0.016$. Honest censoring costs real accuracy, and leaking the answer buys plenty.

8 What this does not do

The score is the additive-model projection of the family history you fed it. Household transmission, dominance, epistasis, indirect genetic effects, and generative assortment are not in A . When they are in the data, they are in μ_i .

Diagnosis and family-history reporting are part of the observation model. Parental-history Alzheimer GWAS is a documented example of survival and participation bias [24]; shallow psychiatric phenotypes can carry heritable confounding [2]. A CIP estimated from an ascertained biobank sample is the wrong $K(\cdot)$. Kaplan–Meier estimates net risk under independent censoring; competing death needs Aalen–Johansen. The mixture (10) inherits both requirements.

Related target families produce related μ_i by construction. A linear GWAS of those scores without a mixed model, or without pruning to non-overlapping families, is the analysis Zhuang et al. [25] warned about.

Three evidence gaps would change how hard Section 7 should be leaned on: a locked vector comparison to LTFHPlus, a real-LD GWAS, and a Gibbs implementation of (10). The liability-dependent onset arm is no longer among them—it is reported above at $\rho = 0.6$ —but a single copula strength is not a sweep, and the pinned encoding’s behaviour under weaker dependence is untested.

Software. `ltpred` is a Python implementation of the construction above (NumPy/SciPy; optional Numba). It does not run a GWAS. Heavier prototypes—sex-limited A , genetic nurture, onset-age decay of r_g , a common-factor summary of r_g , a Monte Carlo EM likelihood—are checkout-only and are not covered here. Source: <https://github.com/bvilhjal/ltpred>. Cite the method actually used [7, 18, 19, 12] and the software.

References

- [1] A. C. Aitken. Note on selection from a multivariate normal population. *Proceedings of the Edinburgh Mathematical Society*, 4(2):106–110, 1935. doi:10.1017/S0013091500008063.

- [2] N. Cai et al. A Perspective on heritable confounding in psychiatric genetics. *Nature Genetics*, 2026. doi:10.1038/s41588-025-02465-y.
- [3] D. S. Falconer. The inheritance of liability to certain diseases, estimated from the incidence among relatives. *Annals of Human Genetics*, 29(1):51–76, 1965.
- [4] L. N. Hazel. The genetic basis for constructing selection indexes. *Genetics*, 28(6):476–490, 1943.
- [5] C. R. Henderson. Best linear unbiased estimation and prediction under a selection model. *Biometrics*, 31(2):423–447, 1975.
- [6] C. R. Henderson. A simple method for computing the inverse of a numerator relationship matrix used in prediction of breeding values. *Biometrics*, 32(1):69–83, 1976.
- [7] M. L. A. Hujoel, S. Gazal, P.-R. Loh, N. Patterson, and A. L. Price. Liability threshold modeling of case–control status and family history of disease increases association power. *Nature Genetics*, 52(5):541–547, 2020. doi:10.1038/s41588-020-0613-6.
- [8] M. L. A. Hujoel, P.-R. Loh, B. M. Neale, and A. L. Price. Incorporating family history of disease improves polygenic risk scores in diverse populations. *Cell Genomics*, 2(7):100152, 2022. doi:10.1016/j.xgen.2022.100152.
- [9] G. L. Jones, M. Haran, B. S. Caffo, and R. Neath. Fixed-width output analysis for Markov chain Monte Carlo. *Journal of the American Statistical Association*, 101(476):1537–1547, 2006.
- [10] K. S. Kendler, H. Ohlsson, J. Sundquist, and K. Sundquist. The family genetic risk score. *JAMA Psychiatry*, 78(6):637–643, 2021. doi:10.1001/jamapsychiatry.2021.0336.
- [11] J. H. Kotecha and P. M. Djurić. Gibbs sampling approach for generation of truncated multivariate Gaussian random variables. In *ICASSP*, 1999.
- [12] M. Dybdahl Krebs et al. Genetic liability estimated from large-scale family data improves genetic prediction, risk score profiling, and gene mapping for major depression. *The American Journal of Human Genetics*, 111(11):2494–2509, 2024. doi:10.1016/j.ajhg.2024.09.009.
- [13] M. Dybdahl Krebs et al. The relationship between genotype- and phenotype-based estimates of genetic liability to psychiatric disorders, in practice and in theory. *The American Journal of Human Genetics*, 113(1):184–201, 2026. doi:10.1016/j.ajhg.2025.11.016.
- [14] S. H. Lee, N. R. Wray, M. E. Goddard, and P. M. Visscher. Estimating missing heritability for disease from genome-wide association studies. *The American Journal of Human Genetics*, 88(3):294–305, 2011.
- [15] J. Z. Liu, Y. Erlich, and J. K. Pickrell. Case–control association mapping by proxy using family history of disease. *Nature Genetics*, 49(3):325–331, 2017. doi:10.1038/ng.3766.
- [16] N. R. Mendell and R. C. Elston. Multifactorial qualitative traits: genetic analysis and prediction of recurrence risks. *Biometrics*, 30(1):41–57, 1974.
- [17] K. Pearson. Mathematical contributions to the theory of evolution. XI. On the influence of natural selection on the variability and correlation of organs. *Philosophical Transactions of the Royal Society A*, 200:1–66, 1903.
- [18] E. M. Pedersen et al. Accounting for age of onset and family history improves power in genome-wide association studies. *The American Journal of Human Genetics*, 109(3):417–432, 2022. doi:10.1016/j.ajhg.2022.01.009.

- [19] E. M. Pedersen et al. ADuLT: an efficient and robust time-to-event GWAS. *Nature Communications*, 14:5553, 2023. doi:10.1038/s41467-023-41210-z.
- [20] M. R. Robinson et al. Genetic evidence of assortative mating in humans. *Nature Human Behaviour*, 1:0016, 2017.
- [21] P. Sham. *Statistics in Human Genetics*. Arnold, London, 1998.
- [22] H.-C. So, J. S. H. Kwan, S. S. Cherny, and P. C. Sham. Risk prediction of complex diseases from family history and known susceptibility loci, with applications for cancer screening. *The American Journal of Human Genetics*, 88(5):548–565, 2011.
- [23] S. Wright. An analysis of variability in number of digits in an inbred strain of Guinea pigs. *Genetics*, 19(6):506–536, 1934.
- [24] Y. Wu et al. Pervasive biases in proxy genome-wide association studies based on parental history of Alzheimer’s disease. *Nature Genetics*, 56:2696–2703, 2024. doi:10.1038/s41588-024-01963-9.
- [25] Z. Zhuang et al. Relatedness, case-control imbalance and tail behaviour in family-history phenotypes. *Bioinformatics*, 2022. doi:10.1093/bioinformatics/btac459.